with the abstract equations

$$\sigma_z|u\rangle = |u\rangle$$

$$\sigma_z|d\rangle = -|d\rangle. \qquad (3.12)$$

- Principle 3: States |*u*> and |*d*> are orthogonal to each other. This can be expressed as

$$\langle u|d\rangle = 0. \qquad (3.13)$$

Recalling our column representations of |*u*> and |*d*> from Eqs. 2.11 and 2.12, we can write Eqs. 3.12 in matrix form as

$$\begin{pmatrix} (\sigma_z)_{11} & (\sigma_z)_{12} \\ (\sigma_z)_{21} & (\sigma_z)_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad (3.14)$$

and

$$\begin{pmatrix} (\sigma_z)_{11} & (\sigma_z)_{12} \\ (\sigma_z)_{21} & (\sigma_z)_{22} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\begin{pmatrix} 0 \\ 1 \end{pmatrix}. \qquad (3.15)$$

There is only one matrix that satisfies these equations. I leave it as an exercise to prove

$$\begin{pmatrix} (\sigma_z)_{11} & (\sigma_z)_{12} \\ (\sigma_z)_{21} & (\sigma_z)_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad (3.16)$$

or, more concisely,